

DHMC Familial Cancer Program
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Result:

Invitae's Multi-Cancer Panel with Preliminary-evidence Pancreatic Cancer Genes was **positive for a mutation in the ATM gene, specifically c.7388_7389ins? (p.Leu2463fs).**

The following 86 genes were evaluated for sequence changes and exonic deletions/duplications: AIP, ALK, APC, ATM, AXIN2, BAP1, BARD1, BLM, BMPR1A, BRCA1, BRCA2, BRIP1, CASR, CDC73, CDH1, CDK4, CDKN1B, CDKN1C, CDKN2A (p14ARF), CDKN2A (p16INK4A), CEBPA, CHEK2, CTNNA1, DICER1, DIS3L2, EGFR, EPCAM (Deletion/duplication testing only), FANCC, FH, FLCN, GATA2, GPC3, GREM1 (Promoter region deletion/duplication testing only), HOXB13, HRAS, KIT, MAX, MEN1, MET, MTF, (c.952G>A, P.Glu318Lys variant only), MLH1, MSH2, MSH3, MSH6, MUTYH, NBN, NF1, NF2, NTHL1, PALB2, PALLD, PDGFRA, PHOX2B, PMS2, POLD1, POLE, POT1, PRKAR1A, PTCH1, PTEN, RAD50, RAD51C, RAD51D, RB1, RECQL4, RET, RUNX1, SDHA, SDHAF2, SDHB, SDHC, SDHD, SMAD4, SMARCA4, SMARCB1, SMARCE1, STK11, SUFU, TERC, TERT, TMEM127, TP53, TSC1, TSC2, VHL, WRN, and WT1.

A variant of uncertain significance (VUS) in the RAD51C gene, specifically c.428A>G (p.Gln143Arg), was detected.

Interpretation:

The most significant consequences of carrying one non-working copy of the ATM gene are increased risks for breast, ovarian and pancreatic cancer. For a female who has not had cancer, lifetime risk of breast cancer is approximately 15-40%, compared to a ~12% lifetime risk for the general US population. The risk for a second, primary breast cancer has not been established. This risk is about 6% in the general population. With an ATM mutation it may be up to twice this (10-12%). Studies indicate that at least for some ATM mutations there is an increased risk for contralateral breast cancer related to radiation scatter during radiation of the affected breast.

The absolute risk for ovarian cancer is less than 3% compared to the general population's risk of about 1.5%. The risk for pancreatic cancer is approximately 5-10% compared to about 1-2% in the general population. The risk for prostate cancer may be increased but this is not certain. It is important to keep in mind that not all individuals who inherit a mutation in the ATM gene will develop cancer.

We reviewed dominant inheritance. With this type of inheritance you only need to inherit one copy of the altered gene to have an increased risk for cancer. Since we all have 2 copies of each gene, one from each parent, there is a 50% chance for each child of a parent who has the mutation to inherit the normal copy of the gene and not be at increased risk for the cancer running in the family. There is also a 50% chance for each child to inherit the altered copy of the gene and therefore be at increased risk for cancer.

- Her daughter has a 50% (1/2) chance of having inherited the altered gene. They should be offered genetic testing to learn whether they are at increased risk.
- Each of her siblings has a 50% chance of having inherited the altered gene and should consider testing.
- Based on family history, it is not clear which side of the family this was likely inherited from. We would suggest that her relatives on the maternal and paternal side of the family be offered testing.

We are happy to see family members here for genetic counseling and testing. Our scheduling secretary can be reached at 603-653-3541. Family members can also go to the website of the National Society of Genetic Counselors at www.nsgc.org in order to find a counselor in their area.

For any family members who are considering family planning, they should be aware that if an individual inherits two non-working copies of the ATM gene (one from each parent), it can lead to a rare, autosomal recessive, childhood onset neurodegenerative disorder affecting multiple body systems. Difficulty with coordinating movements (ataxia) begin in early childhood, usually before age 5. This condition is called ataxia-telangiectasia. Should any relatives in reproductive age test positive for the ATM mutation, they may want their partner to be tested in order to find out if they are at risk of having a child with ataxia telangiectasia. Reproductive options are also available to those interested in preventing passing this condition to their future children. A consultation with a prenatal genetic counselor may be useful for any relatives who may be concerned about reproductive risks.

Variant of Uncertain Significance

It is unclear at this time whether the RAD51C VUS identified is a cancer-associated mutation or a benign change in the gene with no increased cancer risks. Invitae is continually collecting and analyzing their data, in an effort to reclassify these variants as either cancer-causing mutations or benign changes. It is important to remember that a vast majority of variants of uncertain significance are normal, benign changes in the gene.

We will be contacted by the laboratory, in the future, if a reclassification is made and we would then notify her. It is important that her phone number and mailing address stay updated in the Dartmouth-Hitchcock system, in order for us to reach her in the future, should an amended report be issued.

Screening Recommendations for ATM Mutation Carriers:

Breast cancer screening

- Monthly self breast exams and semiannual clinical breast exams
- Annual mammograms with consideration of tomosynthesis starting at age 40.
- Consideration of annual breast MRI screening starting at age 40. Ideally, this should be performed at a facility that has breast MRI directed biopsy capability.
 - Prophylactic, risk-reducing bilateral mastectomy may be considered, although evidence for this recommendation is insufficient

Ovarian cancer screening

- Annual or routine pelvic exams as recommended by a gynecologist or primary care provider

- There is insufficient evidence to recommend risk reducing salpingo-oophorectomy in ATM carriers at this time. Individuals with a family history of ovarian cancer should be managed based on their family history.

Pancreatic cancer screening

Some families with ATM mutations also show an increased incidence of pancreatic cancer, as seen in her and her family. We discussed that two main risk factors in the general population for pancreatic cancer are smoking and heavy drinking. Although there are no clear data on the increased risk of combining smoking and heavy drinking with being an ATM carrier, this test result should be thought of as another good reason, among many others, for ATM mutation carriers not to smoke or drink heavily. We also discussed that there is no established screening for pancreatic cancer and that current expert opinion only recommends screening for pancreatic cancer in ATM carriers with a 1st or 2nd degree relative with pancreatic cancer. For this reason, if her daughter were to test positive we would recommend a consultation with gastroenterology to review current information on pancreatic cancer screening.

Family members interested in testing should be aware of an online support organization called FORCE: Facing Our Risk of Cancer Empowered which may help with assimilating to these genetic test result. In addition, the ClinicalTrials.gov website will have the most up-to-date information of clinical trials available to those who have a gene mutation.